

Independent University, Bangladesh(IUB)



Project Title:

Predicting Heart Disease Risk Using Machine Learning: A Comparative Study

Submitted To:

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Abstract

Heart disease remains a leading cause of death worldwide. Early screening can save lives by directing high-risk patients to timely care. In this project, I build a simple but practical machine-learning pipeline to predict heart-disease risk using three algorithms: K-Nearest Neighbors (KNN), Decision Tree, and Random Forest. The workflow covers data cleaning, exploratory analysis, feature preparation, model training with cross-validation, and fair, side-by-side comparison using accuracy, precision, recall, F1-score, ROC–AUC, and confusion matrices. I also discuss model behavior, basic error analysis, and what this means for real clinics. Results show that tree-based methods (especially Random Forest) usually perform best and are more robust to noise, while KNN is easy to understand but sensitive to feature scaling and parameter choice.

1. Introduction

Cardiovascular disease (CVD) places a heavy burden on families and health systems. Doctors already use medical signs like chest pain type, resting blood pressure, cholesterol, and electrocardiogram results to estimate risk. But with even a modest dataset, machine learning can help by finding patterns that are hard to see by eye.

Goal of this study: I compare three well-known classification models KNN, Decision Tree, and Random Forest to see which model best predicts heart-disease risk on a standard tabular dataset. The focus is not only on raw accuracy, but also on balanced performance (recall for the “disease” class matters), and on how practical and interpretable the models are.

Research questions:

- RQ1: After basic cleaning and feature prep, how well can KNN, Decision Tree, and Random Forest predict heart-disease risk?
- RQ2: Which evaluation metrics are most informative for this task (accuracy vs. recall vs. ROC–AUC)?
- RQ3: What do we learn from feature importance and error patterns that could guide real-world use?

2. Data & Features

2.1 Dataset description

The notebook uses a common heart-disease dataset (tabular, one row per patient). Each row has clinical measurements and a binary label: 1 = heart disease present, 0 = no heart disease. Typical features include: age, sex, chest-pain type (cp), resting blood pressure (trestbps), cholesterol (chol), fasting blood sugar (fbs), resting ECG (restecg), maximum heart rate achieved (thalach), exercise induced angina (exang), ST depression (oldpeak), ST slope (slope), number of major vessels (ca), and thalassemia (thal).

2.2 Data quality and preparation

- Missing values: impute median for numeric, mode for categorical.
- Categorical encoding: one-hot for cp, restecg, slope, thal.
- Scaling: standardize numeric features (z-score). Tree models do not require scaling but a common scaler keeps the pipeline consistent.

- Train/test split: stratified 80/20 split to maintain class balance.
- Cross-validation: 5-fold CV on the training set to tune hyperparameters fairly.

3. Methodology

3.1 Overall workflow

- Load data and inspect shape/dtypes
- Clean and encode (impute, one-hot, scale)
- Split data (stratified) into train/test
- Set up three models with sensible parameter grids
- Cross-validated search for each model
- Evaluate on the held-out test set with multiple metrics
- Plot confusion matrices and ROC curves
- Inspect tree-based feature importance and do short error analysis.

3.2 Models and key hyperparameters

KNN: n_neighbors (k), weights (uniform/distance), metric (minkowski).

Strengths: simple

Weaknesses: sensitive to noise and scaling.

Decision Tree: max_depth, min_samples_split/leaf, criterion (gini/entropy).

Strengths: interpretable

Weaknesses: can overfit.

Random Forest: n_estimators, max_depth, max_features, min_samples_split/leaf, bootstrap.

Strengths: robust, strong default performance

Weaknesses: less transparent than a single tree.

3.3 Evaluation metrics

Accuracy overall correctness; Precision among predicted positives, how many are true; Recall among true positives, how many we caught; F1 – balance of precision and recall; ROC-AUC threshold-independent separability; Confusion matrix – detailed error counts.

4. Exploratory Data Analysis (EDA)

4.1 Class balance

Check counts of positive/negative labels. If imbalance is high, consider class weights or resampling (e.g., SMOTE).

4.2 Univariate and bivariate views

Plot histograms for age, trestbps, chol, thalach, oldpeak; and box/violin plots split by target. Compute correlations among numeric features to detect redundancy.

4.3 Early signals

Generally, higher oldpeak and exang=1 align with disease; higher thalach aligns with no disease; cp and slope tend to be strong predictors. Exact values depend on the dataset instance.

5. Experiments

5.1 Train/validation strategy

Use a stratified 80/20 split. Apply 5-fold cross-validation on the training set. Example grids:

- KNN: $k \in \{3,5,7,9,11,13\}$, $\text{weights} \in \{\text{uniform, distance}\}$
- Decision Tree: $\text{max_depth} \in \{3,5,7,9,\text{None}\}$, $\text{min_samples_leaf} \in \{1,2,4\}$, $\text{criterion} \in \{\text{gini, entropy}\}$
- Random Forest: $\text{n_estimators} \in \{100,200,300\}$, $\text{max_depth} \in \{5,7,9,\text{None}\}$, $\text{min_samples_leaf} \in \{1,2,4\}$, $\text{max_features} \in \{\text{sqrt, log2}\}$

5.2 Final models (example)

KNN best with $k=7$, $\text{weights}='distance'$; Decision Tree best with $\text{max_depth}=5$, $\text{min_samples_leaf}=2$; Random Forest best with $\text{n_estimators}=200$, $\text{max_depth}=\text{None}$, $\text{min_samples_leaf}=1$, $\text{max_features}='sqrt'$. Replace with your notebook's actual best parameters.

6. Discussion

6.1 Which model is best here?

For screening, Random Forest tends to offer the best trade-off between accuracy and recall with strong ROC-AUC. A pruned Decision Tree is more interpretable but may sacrifice recall. KNN is simple but sensitive to scaling.

7. Limitations & Risks

Limitations and risks span both technical and practical aspects. First, data quality and size can constrain what the model learns: missing values, measurement errors, or subtle coding differences between hospitals may reduce accuracy and lead to unstable results. Small datasets also increase the chance of overfitting and make performance estimates overly optimistic. Second, models may not transfer well across settings (dataset shift). A classifier trained on one hospital's population, equipment, or clinical workflow can underperform in another where demographics, prevalence, or testing practices differ. Third, class imbalance is common in medical data and can inflate accuracy while hiding poor recall on the positive (disease) class; careful metric selection and threshold tuning are essential. Fourth, feature leakage and preprocessing mistakes (fitting scalers on the full dataset, using post-diagnosis variables) can inflate validation scores but fail in real use. Fifth, fairness and representation matter: if some subgroups (e.g., women, older adults, or under-served communities) are under-represented, the model may be biased and systematically less accurate for them. Sixth, interpretability varies by model: a single decision tree is easy to explain but may be less accurate. Random Forest is stronger but harder to justify to clinicians and patients without added tools (feature importance, partial dependence, SHAP). Seventh, calibration issues can mislead clinicians if predicted probabilities do not match real risks; probability calibration may be required. Finally, choosing an operating threshold is not trivial: clinical costs of false negatives (missed disease) versus false positives (unnecessary follow-up) differ by context, and thresholds should be set with domain experts and revisited over time.

8. Ethical, Legal, and Privacy Considerations

Ethical, legal, and privacy considerations are central when using machine learning in healthcare. Patient data must be collected and processed with proper consent, legal basis, and strict privacy protection, including de-identification or pseudonymization where appropriate, secure storage, access control, and clear data-retention policies. Use the least data necessary to meet the clinical goal, document why each feature is included, and avoid variables that encode sensitive or protected attributes unless there is a clear clinical justification and governance approval. Provide meaningful explanations of model behavior to clinicians and, when feasible, to patients so decisions remain transparent and contestable. Keep a human-in-the-loop: the model

should assist, not replace, clinical judgment, and clinicians must be empowered to override predictions. Establish governance: version the model, log training data sources and preprocessing steps, record hyperparameters, and maintain an audit trail of changes. Monitor performance and fairness after deployment (including subgroup metrics), and have a clear process for handling complaints, updating thresholds, retraining on fresh data, and rolling back models that drift. Ensure compliance with applicable health-data regulations in your jurisdiction and coordinate with your institution's ethics board or data protection office before deployment.

9. Conclusion

Comparing KNN, Decision Tree, and Random Forest for heart-disease prediction shows Random Forest is generally strongest especially on recall and ROC-AUC, which matter in screening. A pruned Decision Tree remains a useful explainable baseline, while KNN requires careful scaling and parameter tuning. Recommendation: deploy Random Forest with regular monitoring and clear documentation.

Future work: add probability calibration, try gradient boosting (XGBoost/LightGBM), use SHAP for per-patient explanations, and validate on external datasets.